

When Optimal Transport Meets Photo-Realistic Image Dehazing With Unpaired Training

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Abstract—Image hazing is crucial for enhancing the image visibility and mitigating the weather degradations. However, most existing approaches rely on the paired hazy and clean images, which are challenging to obtain in real-world scenarios. To this end, we propose an oriented Bayesian-regularized consistent optimal transport (OBCOT) framework, which formulates the unpaired image dehazing task as an optimal transport (OT) problem. Specifically, we introduce a structure-preserving transport cost, incorporating the structural similarity (SSIM) constraint to minimize the duality gap between the primal and dual formulations, while preserving the structural details of reconstructed images. Furthermore, we derive the Bayesian frequency-domain regularization (BFR) to balance the spectral consistency with clean References and repulsion from hazy patterns. In addition, we employ a pretrained one-step stable diffusion model as the restoration network, which is fine-tuned using the low-rank adaptation (LoRA) adapters and zero convolutional layers, while integrating the domain-specific text prompts for both degraded and clean images to guide the generation process. Extensive experiments demonstrate that our method surpasses the existing well-performing unpaired learning approaches, achieving notable improvements in both the fidelity and photo-realism.

Index Terms—Diffusion model, frequency regularization, image dehazing, optimal transport (OT), unpaired learning.

I. INTRODUCTION

IMAGES captured under hazy conditions often exhibit reduced brightness and diminished contrast, significantly compromising visual quality and impeding their utility in downstream vision-based tasks, such as object detection and instance segmentation [1], [2], [3], [4], [5]. Hazy conditions frequently lead to not only color distortion and contrast reduction but also blurring effects, particularly in dense or nonhomogeneous scenarios [6], [7], [8], [9]. Early approaches [10], [11], [12] predominantly relied on physical priors to estimate the necessary atmospheric parameters before restoring the latent clear images. While these methods demonstrate effective dehazing performance, they are prone to over-dehazing due to the inherent limitations of hand-crafted priors, which are often scenario-dependent. Recent advances [8], [13],

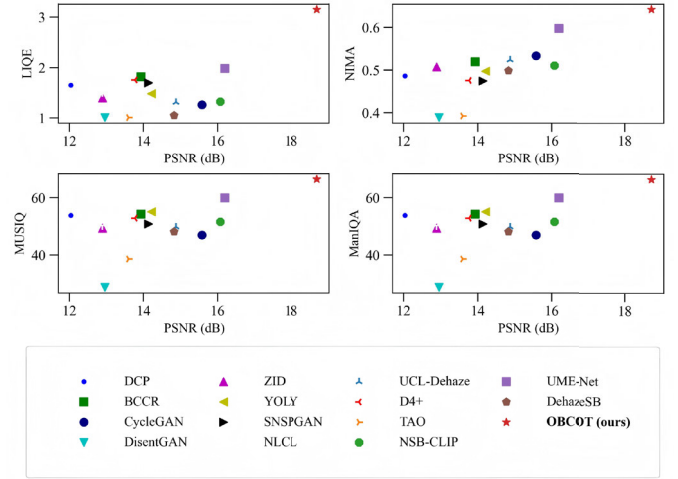


Fig. 1. Intuitive comparisons of our proposed method and the other existing approaches between the average PSNR and four image naturalness assessment metrics. Our model achieves the currently best performance in the supervised protocols, while preserving the significant improved naturalness.

[14], [15], [16], [17], [18] in deep learning have prompted a paradigm shift toward data-driven dehazing techniques. These methods typically employ synthetically generated paired datasets for training, given the challenges associated with acquiring real-world hazy-clear image pairs. Although they achieve improved performance, such paired training frameworks frequently exhibit limited generalization in real-world settings, primarily due to the suboptimal performance of networks trained on synthetic data that lack authentic haze characteristics. To address the constraints of paired training, recent studies [19], [20], [21], [22] have explored unpaired training paradigms for dehazing, eliminating the need for precisely matched data pairs.

However, precisely learning the mapping between hazy and clean domains without relying on paired data remains a significant challenge. Although existing domain adaptation methods [22], [23] have achieved some progress, the dehazed images often still exhibit noticeable haze degradation. While several methods [24], [25] have explicitly addressed blur removal in challenging real-world environments, our current work focuses on a more general formulation of haze-induced degradation. In domain translation, optimal transport (OT) establishes a mapping between two probability distributions by minimizing the transportation cost, with its core idea being the search for the most efficient allocation scheme. Meanwhile, unpaired

Received 12 August 2025; revised 3 December 2025; accepted 9 March 2026. Date of publication 9 April 2026; date of current version 2 September 2026. This work was supported by the National Natural Science Foundation of China under Grant 52572353 and Grant 52441205. (Corresponding author: Tao Gao.)

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Digital Object Identifier 10.1109/TNNLS.2026.3673760

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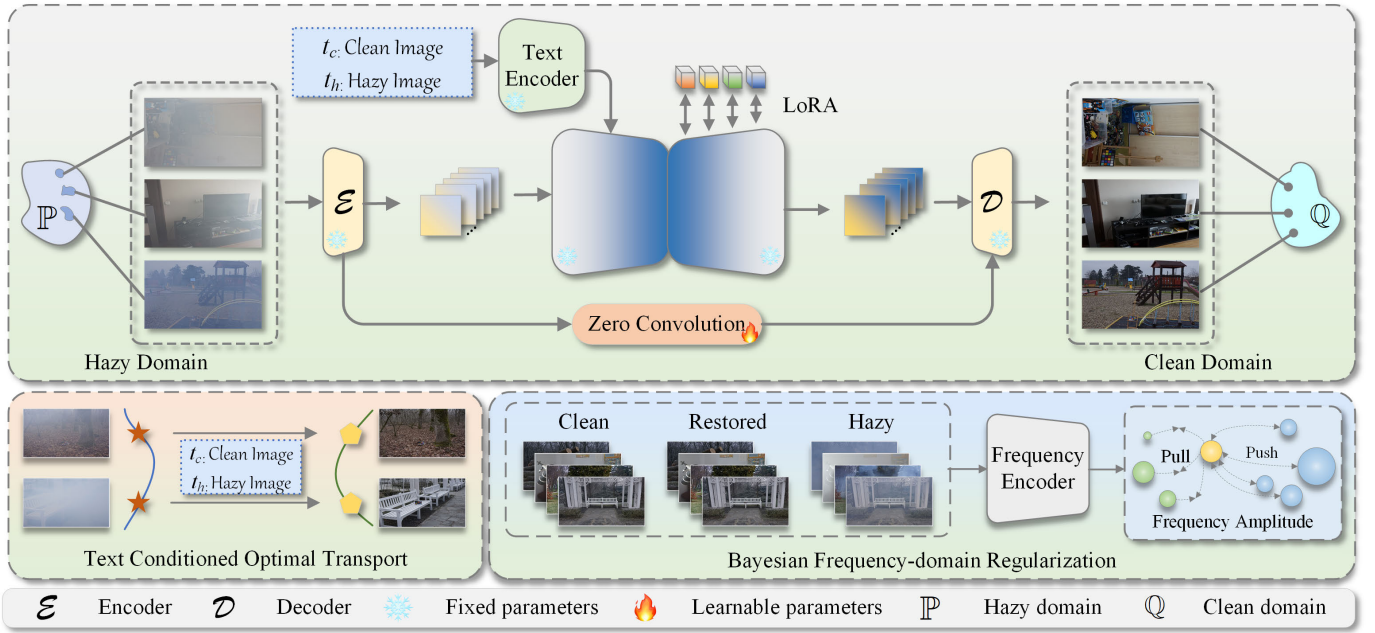


Fig. 2. Overview of our proposed OBCOT for unpaired haze removal. We formulate the unpaired restoration task as an OT problem, seeking an optimal mapping to simultaneously eliminate degradations while preserving structural integrity and frequency-domain consistency.

image dehazing aligns the distribution of hazy images (source domain) with that of clear images (target domain) without requiring strictly paired data. Obviously, OT aligns well with the learning objective of unpaired dehazing, which implicitly learns the domain transformation and reconstructs high-quality dehazed images from hazy inputs. Another critical challenge in unpaired dehazing is ensuring the structural consistency and frequency regularization between hazy and dehazed images to suppress the misaligned information caused by the unpaired learning framework. While CycleGAN [19], [26], [27] serves as a classical solution and has been widely adopted in previous unpaired dehazing researches, it necessitates additional training of generators and discriminators, incurring the extra computational overhead. Therefore, our work focuses on leveraging OT between hazy and unpaired clean images while preserving structural integrity and frequency accuracy during domain transformation.

In this work, we formulate the unpaired haze removal as an OT problem, proposing an oriented Bayesian-regularized consistent OT (OBCOT) framework. Specifically, we first cast OT as a Lagrange optimization problem, enabling its solution via adversarial learning [28], [29], [30]. To preserve the structural integrity, we introduce a structure-preserving transport cost that exploits the local quasi-convexity of structural similarity (SSIM), minimizing the duality gap between primal and dual formulations and ensuring weak duality holds [31]. Unlike existing methods [32], [33], our framework employs a pretrained one-step stable diffusion model as the restoration backbone, incorporating domain-specific text prompts for both degraded and clean images, guiding reconstruction through human-specified cues. As shown in Fig. 2, the pretrained model is adapted via low-rank adaptation (LoRA) fine-tuning

and zero convolutional layers, ensuring compatibility with pretrained weights while maintaining computational stability.

Furthermore, we analyze the frequency-domain characteristics of real-world haze, observing that it introduces a strong low-frequency bias due to atmospheric scattering, while attenuating mid-to-high frequency components corresponding to texture and edge information. This spectral shift creates a misalignment between hazy and clean image distributions. Existing adversarial methods [23], [26], [27], [32] often fail to enforce cross-spectral consistency, as their discriminators operate primarily on spatial features and lack explicit frequency-domain constraints. This can result in either over-smoothed outputs that lack high-frequency details or images with residual haze artifacts. To this end, we propose the Bayesian frequency-domain regularization (BFR), which formulates the unpaired dehazing task as a maximum a posteriori (MAP) estimation problem in the frequency domain. Unlike GAN discriminators, our MAP-based approach explicitly models a spectral likelihood to attract the output toward clean references and a contrastive prior to repel hazy patterns. This provides a principled and interpretable framework for ensuring spectral fidelity and suppressing degradation without the instability often associated with adversarial training. Therefore, as Fig. 1 shown, our OT formulation not only enhances structural consistency but also preserves the frequency alignment, preventing excessive distortion of critical details while maintaining photo-realism.

Fig. 2 depicts the overall framework of our proposed OBCOT for photo-realistic unpaired image dehazing. The main contributions of our work are as follows.

- 1) We establish a theoretical formulation of image dehazing as an OT problem. Subsequently, this OT

formulation is reformulated within a Lagrangian optimization framework, thereby enabling its implementation via a sequence of tractable optimization objectives.

- 2) We introduce a structure-preserving transport cost functional that minimizes the duality gap between the primal and dual OT formulations.
- 3) We propose a text-oriented pretraining paradigm for the transport network, which utilizes the textual prompts corresponding to hazy and clean domains to regulate the generative process.
- 4) We derive the BFR that enforces cross-frequency distributional consistency between the dehazed and clean image distributions.

II. RELATED WORK

A. Unpaired Image Dehazing

With the development of deep learning [7], [9], [17], [34], [35], some efforts concentrate on designing networks to estimate the parameters of the atmospheric scattering model or directly learn the mapping from hazy to clear images. DehazeNet [36] is the pioneer in estimating transmission maps in an end-to-end manner. C2PNet [14] customizes a curriculum learning to improve dehazing performance. Despite the existence of various proposed networks, their dehazing performance on real-world images remains limited since they are trained on synthetic datasets. To address the bottleneck of paired training, recent studies have focused on unpaired training strategies to improve the model's generalization capabilities. For example, RefineDNet [37] integrates the dark channel prior into a two-stage dehazing network for unpaired learning. D4 + [21] decomposes the atmospheric scattering model and develops a self-augmented dehazing framework built on CycleGAN [27]. UCL-Dehaze [22] proposes an unpaired contrastive learning paradigm for image dehazing. UME-Net [19] proposes an unsupervised multibranch with high-frequency enhancement network for restoring edge and texture information in obscured by dense haze. These methods all employ GANs to generate dehazed images. Despite achieving better performance on real-world hazy images, they still fail to produce satisfactory dehazing results due to the limited transmission mapping capability of the generator.

B. Ot in Image Restoration

Recent advances in deep learning have driven significant progress in image restoration [8], [9], [38], [39], [40]. Concurrently, OT theory, which seeks an ideal mapping or plan to minimize the cost of transforming one distribution into another [41], [42], has been increasingly applied to image translation tasks [28], [29], [43]. For instance, Korotin et al. [44] employ the neural networks within a duality framework to compute the transport maps for unpaired image translation. Gu et al. [29] utilize the OT to guide conditional score-based diffusion models for super-resolution, while [45] relaxes the transport constraint in the Monge formulation via a W_1 discrepancy penalty, ensuring distributional alignment. A prominent research direction involves using OT to directly enhance restoration quality. Adrai et al. [46] demonstrate that

OT could improve both the perceptual quality and fidelity of reconstructed images without requiring retraining. Building on the concept of iterative refinement, Tang et al. [43] propose a two-pass transport map framework, where an initial model estimates a transport residual that guides a secondary restoration process. However, this approach relies on partially paired training data and requires two full model executions for reconstruction. Furthermore, it necessitates retraining all network parameters, which limits its efficiency and scalability. Most recently, Vasa et al. [47] introduce a context-informed OT learning framework for unpaired fundus image enhancement. This paradigm preserves the local structures and minimizes the artifacts compared to standard generative methods [26], [27], which often struggle with contextual information.

III. PROPOSED METHOD

In this section, we introduce our proposed method from the image dehazing formulation, the structure-preserving transport cost, the one-step text-oriented pretraining model, and the BFR.

A. Image Dehazing Formulation

Image dehazing addresses the inverse problem of recovering a clean image from a hazy observation by mitigating atmospheric degradation while preserving the underlying structure. Formally, this can be cast as learning a restoration operator that maps from the space of degraded observations to the space of clean images. Let the clean image space be denoted by $(\mathcal{Y}, p_c)$ and the hazy image space by $(\mathcal{X}, p_h)$, where p_c and p_h represent the probability distributions over clean and hazy images, respectively. The restoration process is then characterized by a measurable transport map $T : \mathcal{X} \rightarrow \mathcal{Y}$.

Definition 1: Given two probability spaces $(\mathcal{X}, p_h)$ and $(\mathcal{Y}, p_c)$ representing hazy and clean image domains with unpaired samples, the objective of unpaired image dehazing is to identify a transport map $T : \mathcal{X} \rightarrow \mathcal{Y}$ that satisfies the distributional alignment condition

$$T_{\#}p_h = p_c \quad (1)$$

where $T_{\#}$ denotes the push-forward measure, while minimizing an expected distortion functional $\mathbb{E}_{\mathbf{x} \sim p_h}[d(\mathbf{x}, T(\mathbf{x}))]$ for a suitable perceptual or pixel-wise metric d .

The optimal restoration criterion generally encompasses three competing objectives [29], [43], [48], [49]; 1) effective suppression of atmospheric scattering artifacts; 2) maximal preservation of the intrinsic image content; and 3) reconstruction that conforms to the statistical properties of natural images. Considering the Markov chain $\mathcal{Y} \rightarrow \mathcal{X} \rightarrow \hat{\mathcal{X}}$, the data processing inequality implies

$$\mathcal{I}(\hat{\mathcal{X}}; \mathcal{X}) \leq \mathcal{I}(\mathcal{Y}; \mathcal{X}) \quad (2)$$

where $\mathcal{I}(\cdot; \cdot)$ denotes mutual information. Consequently, an information-theoretic formulation that balances distortion and perceptual quality can be expressed as

$$\max_T \mathcal{I}(T(\mathcal{X}); \mathcal{Y}) \quad \text{s.t.} \quad d(p_{\hat{\mathcal{X}}}, p_c) \leq 0 \quad (3)$$

where $d(\cdot, \cdot)$ is a statistical divergence measuring the discrepancy between two probability distributions, satisfying $d(p, q) \geq 0$ and $d(p, q) = 0 \iff p = q$. The constraint $d(p_{\mathcal{X}}, p_{\mathcal{Y}}) \leq 0$ enforces perfect perceptual quality by ensuring the output distribution matches that of natural clean images.

In the absence of paired supervision, we reformulate the dehazing objective as an OT problem. Specifically, we seek a transport map T^* that minimizes a structure-aware transport cost while aligning the hazy and clean distributions, thereby simultaneously addressing the dual objectives of distortion minimization and perceptual quality enhancement without requiring paired supervision. The selection of an appropriate discrepancy measure $d(\cdot, \cdot)$ to enforce $T_{\#}p_h = p_c$ is crucial. While various probability metrics exist, OT provides a theoretically grounded framework particularly suited to unpaired image restoration. Unlike f -divergences [Kullback–Leibler (KL) and Jensen–Shannon (JS)], which suffer from vanishing gradients when distributions have disjoint supports, or kernel-based measures like maximum mean discrepancy (MMD), whose effectiveness depends heavily on kernel selection, OT incorporates the underlying geometry of the data space through the cost function. This geometric sensitivity ensures that minimizing the Wasserstein distance not only aligns distributions but also preserves structural relationships during domain transformation, a property essential for maintaining image fidelity.

B. Structure-Preserving Transport Cost

Let $\mathcal{X}$ denote the domain of hazy images and $\mathcal{Y}$ the domain of clean images. Their associated probability distributions are denoted by $\mathbb{P} \in \mathcal{P}(\mathcal{X})$ and $\mathbb{Q} \in \mathcal{P}(\mathcal{Y})$, respectively. The objective of unpaired image dehazing is to learn a mapping that transports the distribution $\mathbb{P}$ to $\mathbb{Q}$ while preserving the structural content. The formulation of OT can be given as follows.

Definition 2: $T : \mathcal{X} \rightarrow \mathcal{Y}$ is a transport map that transports the probability measure $\mathbb{P} \in \mathcal{P}(\mathcal{X})$ to $\mathbb{Q} \in \mathcal{P}(\mathcal{Y})$, if

$$\mathbb{Q}(A) = \mathbb{P}(T^{-1}(A)) \quad (4)$$

for any $\mathbb{Q}$ -measurable sets A . Equivalently, for any integrable function $\varphi : \mathcal{Y} \rightarrow \mathbb{R}$

$$\int_{\mathcal{Y}} \varphi(\mathbf{y}) d\mathbb{Q}(\mathbf{y}) = \int_{\mathcal{X}} \varphi(T(\mathbf{x})) d\mathbb{P}(\mathbf{x}). \quad (5)$$

The classical Monge formulation [41] of OT seeks a transport map that minimizes the expected transportation cost.

Definition 3: Let $c : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ be a continuous cost function. Given two probability measures $\mathbb{P} \in \mathcal{P}(\mathcal{X})$ and $\mathbb{Q} \in \mathcal{P}(\mathcal{Y})$, the Monge problem is to find a transport map $T : \mathcal{X} \rightarrow \mathcal{Y}$ satisfying $T_{\#}\mathbb{P} = \mathbb{Q}$ that minimizes

$$\inf_{T: T_{\#}\mathbb{P}=\mathbb{Q}} \int_{\mathcal{X}} c(\mathbf{x}, T(\mathbf{x})) d\mathbb{P}(\mathbf{x}). \quad (6)$$

A minimizer T^* , if it exists, is called an OT map.

Since the existence of such a transport map is not guaranteed under general conditions, the relaxed Kantorovich formulation [42] considers transport plans.

Definition 4: Let $\Gamma(\mathbb{P}, \mathbb{Q})$ denote the set of all joint probability measures on $\mathcal{X} \times \mathcal{Y}$ with marginals $\mathbb{P}$ and $\mathbb{Q}$, i.e.,

$$\Gamma(\mathbb{P}, \mathbb{Q}) = \{\gamma \in \mathcal{P}(\mathcal{X} \times \mathcal{Y}) | \pi_{\mathcal{X}}^* \gamma = \mathbb{P}, \pi_{\mathcal{Y}}^* \gamma = \mathbb{Q}\} \quad (7)$$

where π^1 and π^2 are the canonical projections. The Kantorovich problem is to find a coupling $\gamma \in \Gamma(\mathbb{P}, \mathbb{Q})$ that minimizes

$$\inf_{\gamma \in \Gamma(\mathbb{P}, \mathbb{Q})} \int_{\mathcal{X} \times \mathcal{Y}} c(\mathbf{x}, \mathbf{y}) d\gamma(\mathbf{x}, \mathbf{y}). \quad (8)$$

A minimizer γ^* always exists under mild conditions and is called an OT plan.

Theorem 1: For two probability measures $\mathbb{P} \in \mathcal{P}(\mathcal{X})$ and $\mathbb{Q} \in \mathcal{P}(\mathcal{Y})$ and a continuous cost function $c : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, the following duality holds:

$$\begin{aligned} & \inf_{\gamma \in \Gamma(\mathbb{P}, \mathbb{Q})} \int_{\mathcal{X} \times \mathcal{Y}} c(\mathbf{x}, \mathbf{y}) d\gamma(\mathbf{x}, \mathbf{y}) \\ &= \sup_{\substack{(\varphi, \psi) \in L^1(\mathbb{P}) \times L^1(\mathbb{Q}) \\ \varphi(\mathbf{x}) + \psi(\mathbf{y}) \leq c(\mathbf{x}, \mathbf{y})}} \left(\int_{\mathcal{X}} \varphi(\mathbf{x}) d\mathbb{P}(\mathbf{x}) + \int_{\mathcal{Y}} \psi(\mathbf{y}) d\mathbb{Q}(\mathbf{y}) \right) \end{aligned} \quad (9)$$

where φ and ψ are real-valued measurable functions often referred to as Kantorovich potentials.

The Kantorovich–Rubinstein duality theorem [50] provides a powerful theoretical foundation for our approach, offering several crucial advantages. By transforming the primal linear programming problem into a concave maximization over Kantorovich potentials (φ, ψ) , it enhances computational tractability for gradient-based optimization. Furthermore, when the cost function c is a metric, the dual formulation specializes to the 1-Wasserstein distance $W_1(\mathbb{P}, \mathbb{Q}) = \sup_{\|\varphi\|_L \leq 1} (\int \varphi d\mathbb{P} - \int \varphi d\mathbb{Q})$, which establishes a direct connection to adversarial training paradigms. This dual perspective not only ensures the existence and regularity of OT plans but also provides a stable and interpretable framework for distribution alignment, addressing the instability often encountered in conventional unpaired learning methods.

These formulations provide a principled framework for unpaired domain translation. In the context of unpaired dehazing, the objective is to find a mapping from the hazy domain $\mathcal{X}$ to the clean domain $\mathcal{Y}$ that minimizes a perceptually meaningful distortion while ensuring the transported distribution matches the target distribution $\mathbb{Q}$. By parameterizing the transport map as a deep neural network T_{θ} with parameters θ , we formulate the unpaired dehazing problem as the following constrained optimization by:

$$\min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathbb{P}} [c(\mathbf{x}, T_{\theta}(\mathbf{x}))^{\beta}], \quad \text{s.t. } (T_{\theta})_{\#} \mathbb{P} = \mathbb{Q} \quad (10)$$

where $\beta \geq 1$ and the constraint enforces distributional alignment. To render this problem computationally tractable, we introduce a Lagrange multiplier $\lambda > 0$ and consider the Lagrangian relaxation as

$$\min_{\theta} \mathbb{E}_{\mathbf{x} \sim \mathbb{P}} [c(\mathbf{x}, T_{\theta}(\mathbf{x}))^{\beta}] + \lambda \mathcal{D}((T_{\theta})_{\#} \mathbb{P}, \mathbb{Q}) \quad (11)$$

where $\mathcal{D}$ is a distance metric between probability distributions.

Lemma 1: Any minimizer of the relaxed problem (11) satisfies the constraint $(T_{\theta})_{\#} \mathbb{P} = \mathbb{Q}$ and is thus a solution to the constrained problem (10).

This establishes the equivalence between the original OT problem and its relaxed Lagrangian form for unpaired image dehazing. Furthermore, a critical consideration in unpaired image dehazing is the design of the cost function c . Traditional pixel-wise norms such as ℓ_1 or ℓ_2 may not align well with human perceptual quality. The SSIM [31], which incorporates luminance, contrast, and structure comparisons, offers a perceptually more meaningful measure. We therefore propose a composite transport cost that combines an ℓ_1 fidelity term with a structural dissimilarity term as

$$c(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1 - \text{SSIM}(\mathbf{x}, \mathbf{y}). \quad (12)$$

The negative SSIM term acts as a structural penalty, encouraging the transport map to preserve local image structures. This hybrid cost promotes both pixel-wise accuracy and perceptual naturalness.

Proposition 1: Let $\mathbf{x}, \mathbf{y}$ be image patches within a local window. The function $\phi(\mathbf{x}) = -\text{SSIM}(\mathbf{x}, \mathbf{y})$ is locally quasi-convex in $\mathbf{x}$ for fixed $\mathbf{y}$. Specifically, for any convex combination $\mathbf{z} = m\mathbf{x} + (1 - m)\mathbf{y}$ with $m \in [0, 1]$

$$\text{SSIM}(\mathbf{z}, \mathbf{y}) \geq \min(\text{SSIM}(\mathbf{x}, \mathbf{y}), \text{SSIM}(\mathbf{y}, \mathbf{y})). \quad (13)$$

Consequently, $\phi(\mathbf{z}) \leq \max(\phi(\mathbf{x}), \phi(\mathbf{y})) = \phi(\mathbf{x})$.

The proposed cost function (12) inherits beneficial optimization properties from this quasi-convexity structure: 1) the loss landscape is smoother with reduced presence of spurious local minima; 2) gradient-based updates exhibit improved stability due to the normalized statistical nature of SSIM; and 3) convergence rates are accelerated owing to the regularization effect of the quasi-convex term. This formulation provides a theoretically grounded and practically effective framework for unpaired image dehazing via OT.

C. One-Step Text-Oriented Pretraining Model

Our pipeline employs a pretrained single-step stable diffusion model [51], [52] as the foundational generative prior. In accordance with the findings of [52], which demonstrate that auxiliary conditional encoder branches can induce feature conflicts and degrade sample fidelity in one-step diffusion models, we implement an alternative conditioning paradigm. Rather than introducing supplementary network branches, we perform direct input conditioning by substituting the standard Gaussian noise vector with our input images. To facilitate the adaptation of the foundational model to this modified input modality while preserving its pretrained knowledge, we integrate LoRA modules [53] into a subset of the decoder layers. This architectural modification is schematically illustrated in Fig. 2. Let $\mathcal{T}_c$ and $\mathcal{T}_h$ denote the textual descriptors for the *clean* and *hazy* image domains, respectively. We define the text-conditioned generator T_θ as a parametric mapping by

$$T_\theta : (\mathbf{x}, \mathcal{T}_c) \rightarrow \hat{\mathbf{x}} \quad (14)$$

where $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ represents an input hazy image, and $\hat{\mathbf{x}}$ denotes the corresponding generated dehazed output, which is constrained to reside within the clean image manifold. This formulation establishes that, given a text prompt for the target

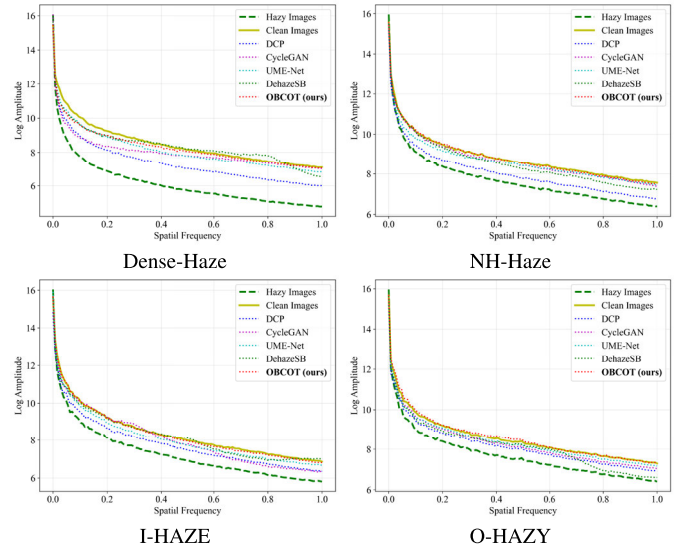


Fig. 3. Spectral analysis of hazy images, clean images, and the results of existing methods. The average amplitude spectra of hazy and clean images depict the low-frequency bias and high-frequency attenuation caused by haze degradation. Our proposed OBCOT effectively restores the spectral profile, closely matching that of the clean image, while existing methods fail to restore high-frequency details or introduce spectral artifacts.

domain ($\mathcal{T}_c$), the generator should learn a transformation that maps samples from the hazy distribution $\mathbb{P}$ toward the clean distribution $\mathbb{Q}$.

However, this objective alone does not ensure the preservation of the input image's underlying content structure, as the model may converge to a trivial solution that ignores the input $\mathbf{x}$ and merely samples from the marginal distribution of $\mathbb{Q}$. To enforce content consistency and prevent unnecessary alterations when the input already belongs to the target domain, we impose bi-directional identity constraints. Formally, these constraints are expressed as

$$\begin{aligned} T_\theta(\mathbf{x}, \mathcal{T}_h) &= \mathbf{x} \quad \text{for } \mathbf{x} \sim \mathbb{P} \\ T_\theta(\mathbf{y}, \mathcal{T}_c) &= \mathbf{y} \quad \text{for } \mathbf{y} \sim \mathbb{Q}. \end{aligned} \quad (15)$$

These constraints ensure that the generator acts as an identity operator when the input image and the conditioning text prompt are domain-aligned. Integrating these terms, the complete objective function derived from (11) is formulated as the minimization of the following loss:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathbb{P}} [c(\mathbf{x}, T_\theta(\mathbf{x}, \mathcal{T}_c))] + \mathbb{E}_{\mathbf{x} \sim \mathbb{P}} [c(\mathbf{x}, T_\theta(\mathbf{x}, \mathcal{T}_h))] \\ &+ \mathbb{E}_{\mathbf{y} \sim \mathbb{Q}} [c(\mathbf{y}, T_\theta(\mathbf{y}, \mathcal{T}_c))] + \lambda \cdot \mathcal{D}((T_\theta)_\# \mathbb{P}, \mathbb{Q}). \end{aligned} \quad (16)$$

The first three terms constitute domain-specific consistency regularizers, while the final term $\mathcal{D}(\cdot, \cdot)$ quantifies the divergence between the transformed hazy distribution $(T_\theta)_\# \mathbb{P}$ and the target clean distribution $\mathbb{Q}$. Critically, the identity regularization terms operate independently of the distribution-matching term $\mathcal{D}(\cdot, \cdot)$ and are not included in its computation.

D. Bayesian Frequency-Domain Regularization

To regularize the spectral characteristics of generated images via their frequency-domain probability distributions,

we cast the unpaired image dehazing problem as an MAP estimation in the frequency domain, introducing a BFR framework. The theoretical motivation stems from a statistical examination of haze-induced degradation in the spectral domain. Let $\mathcal{F}(\mathbf{x})$ and $\mathcal{F}(\mathbf{y})$ denote the Fourier transforms of a hazy image $\mathbf{x}$ and its clean counterpart $\mathbf{y}$, respectively. Empirical analysis indicates that the expected amplitude spectrum of hazy images, $\mathbb{E}[|\mathcal{F}(\mathbf{x})|]$, exhibits two prominent deviations relative to that of clean images (illustrated in Fig. 3): 1) a substantial energy augmentation in the low-frequency band, attributable to the additive-multiplicative amalgam of atmospheric light and scattering; and 2) a marked attenuation of amplitude in the mid-to-high frequency bands, which predominantly encode fine-scale textures, edges, and structural boundaries.

This spectral misalignment is inadequately addressed by prevailing unpaired methodologies. For instance, adversarial frameworks, such as CycleGAN [27] and DehazeSB [54], either fail to fully reconstruct high-frequency components or introduce spectral artifacts in their pursuit of spatial-domain distribution matching, as their discriminative networks lack explicit spectral constraints. This limitation of adversarial training motivates our adoption of an MAP formulation. Whereas a GAN discriminator $\mathcal{D}$ minimizes a distributional divergence $d(\mathcal{Q}(\hat{\mathcal{X}}), \mathcal{Q}(\mathcal{Y}))$ in the spatial domain, it affords no direct guarantee of frequency-domain alignment. Our MAP estimator explicitly decomposes the inference into a spectral likelihood $P(\mathbf{y}|T_\theta(\mathbf{x}))$ and a contrastive prior $P(T_\theta(\mathbf{x})|\mathbf{x})$. This permits independent and direct regulation of attraction toward pristine spectral profiles and repulsion from hazy patterns, thereby fostering more stable optimization and superior spectral fidelity.

Given a hazy image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ and an unpaired clean image $\mathbf{y}$, the objective is to learn a dehazing mapping $T_\theta(\mathbf{x})$ that recovers the underlying scene radiance. This problem is formulated as probabilistically modeling the spectral correlations between $\mathbf{x}$ and $\mathbf{y}$, while explicitly accounting for their statistical discrepancies to steer the model toward faithful restoration. For a hazy input $\mathbf{x}$ and its unpaired clean reference $\mathbf{y}$, the optimal dehazed output $G_\theta^*(\mathbf{x})$ is obtained by maximizing the posterior probability

$$G_\theta^*(\mathbf{x}) = \arg \max_{G_\theta(\mathbf{x})} P(G_\theta(\mathbf{x})|\mathbf{y}, \mathbf{x}) \quad (17)$$

where the joint posterior distribution $P(G_\theta(\mathbf{x})|\mathbf{y}, \mathbf{x})$ factorizes into three constituent components as

$$\underbrace{P(\mathbf{y}|G_\theta(\mathbf{x}))}_{\text{spectral likelihood}} \underbrace{P(G_\theta(\mathbf{x})|\mathbf{x})}_{\text{contrastive prior}} \underbrace{P(G_\theta(\mathbf{x}))}_{\text{spatial prior}}. \quad (18)$$

This probabilistic decomposition provides a principled foundation for implementing each component in the frequency domain. The likelihood term necessitates careful modeling of cross-image spectral relationships, which we address by positing complex Gaussian distributions in the Fourier domain. The corresponding negative log-likelihood reduces to

$$\mathcal{L}_{\text{like}} = \frac{1}{\sigma_l^2} \|\mathcal{F}(G_\theta(\mathbf{x})) - \mathcal{F}(\mathbf{y})\|_F^2 + C \quad (19)$$

where C encapsulates constant terms, $\|\cdot\|_F$ denotes the Frobenius norm, and σ_l^2 is the variance parameter.

While the likelihood term enforces fidelity to clean references, effective dehazing demands explicit modeling and suppression of haze-induced spectral distortions. This motivates our contrastive prior formulation, which actively repels hazy spectral patterns.

Proposition 2: The contrastive prior implements frequency-domain repulsion through the objective

$$\mathcal{L}_{\text{prior}} = -\frac{\lambda_p}{\sigma_p^2} \|\mathcal{F}(G_\theta(\mathbf{x})) - \mathcal{F}(\mathbf{x})\|_F^2 \quad (20)$$

where $\lambda_p > 0$ modulates the repulsion intensity and σ_p^2 determines the spectral bandwidth of influence.

Proof: We model the prior probability density using a Gibbs distribution

$$P(G_\theta(\mathbf{x})|\mathbf{x}) = \frac{1}{Z(\mathbf{x})} \exp(-E(G_\theta(\mathbf{x}), \mathbf{x})) \quad (21)$$

with the partition function

$$Z(\mathbf{x}) = \int \exp(-E(G_\theta(\mathbf{x}), \mathbf{x})) dG_\theta(\mathbf{x}). \quad (22)$$

The energy function

$$E(G_\theta(\mathbf{x}), \mathbf{x}) = \lambda_p \sum_{u,v} \frac{|\mathcal{F}(G_\theta(\mathbf{x}))|_{u,v} - |\mathcal{F}(\mathbf{x})|_{u,v}|^2}{\sigma_p^2} \quad (23)$$

quantifies the spectral divergence between the dehazed and hazy images. The negative log-prior $-\log P(G_\theta(\mathbf{x})|\mathbf{x})$ thus becomes

$$E(G_\theta(\mathbf{x}), \mathbf{x}) + \log Z(\mathbf{x}) \quad (24)$$

$$= \underbrace{\lambda_p \sum_{u,v} \frac{|\mathcal{F}(G_\theta(\mathbf{x}))|_{u,v} - |\mathcal{F}(\mathbf{x})|_{u,v}|^2}{\sigma_p^2}}_{\text{Repulsive Energy}} + C'. \quad (25)$$

For gradient-based optimization, we invert the sign to transform attraction into repulsion, yielding

$$\begin{aligned} \mathcal{L}_{\text{prior}} &= -\lambda_p \sum_{u,v} \frac{|\mathcal{F}(G_\theta(\mathbf{x}))|_{u,v} - |\mathcal{F}(\mathbf{x})|_{u,v}|^2}{\sigma_p^2} \\ &= -\frac{\lambda_p}{\sigma_p^2} \|\mathcal{F}(G_\theta(\mathbf{x})) - \mathcal{F}(\mathbf{x})\|_F^2. \end{aligned} \quad (26)$$

This formulation produces gradient directions that actively separate the spectrum of $G_\theta(\mathbf{x})$ from the haze-corrupted spectrum of $\mathbf{x}$. ■

Definition 5: The spatial prior $P(G_\theta(\mathbf{x}))$ enforces piecewise smoothness through total variation regularization by

$$P(G_\theta(\mathbf{x})) \propto \exp(-TV(G_\theta(\mathbf{x}))) \quad (27)$$

where the total variation seminorm $TV(G_\theta(\mathbf{x}))$ is defined as

$$TV(G_\theta(\mathbf{x})) = \sum_{x,y} \sqrt{|\nabla_x G_\theta(\mathbf{x})(x,y)|^2 + |\nabla_y G_\theta(\mathbf{x})(x,y)|^2}. \quad (28)$$

The corresponding loss term is $\mathcal{L}_{\text{tv}} = TV(G_\theta(\mathbf{x}))$.

Algorithm 1 OBCOT

Input: Unpaired dataset $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$ with $\mathbf{x}_i \sim \mathbb{P}$ (hazy) and $\mathbf{y}_i \sim \mathbb{Q}$ (clean); Text prompts $\mathcal{T}_h$ ('hazy image') and $\mathcal{T}_c$ ('clean image'); Learning rates η_G, η_D ; Batch size b ; Hyper-parameters $\alpha, \beta, \gamma, \lambda_{gp}, \lambda_{ot}$; Number of critic updates per generator step n_{critic} .

Output: Trained restoration network T_θ and critic D_ϕ .

- 1 Initialize θ of T_θ and ϕ of D_ϕ .
- 2 **while** not converged **do**
- 3 **for** $t = 1$ to n_{critic} **do**
- 4 Sample batch $\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^b \sim \mathcal{D}$
- 5 $\hat{\mathbf{x}}_i \leftarrow T_\theta(\mathbf{x}_i, \mathcal{T}_c)$
- 6 $\tilde{\mathbf{y}}_i \leftarrow \epsilon \mathbf{y}_i + (1 - \epsilon) \hat{\mathbf{x}}_i, \epsilon \sim U(0, 1)$
- 7 $\mathcal{L}_{gp} \leftarrow \lambda_{gp} \cdot \frac{1}{b} \sum_{i=1}^b (\|\nabla_{\tilde{\mathbf{y}}_i} D_\phi(\tilde{\mathbf{y}}_i)\|_2 - 1)^2$
- 8 $\mathcal{L}_D \leftarrow \frac{1}{b} \sum_{i=1}^b [D_\phi(\mathbf{y}_i) - D_\phi(\hat{\mathbf{x}}_i)] + \mathcal{L}_{gp}$
- 9 $\phi \leftarrow \phi - \eta_D \cdot \nabla_\phi \mathcal{L}_D$
- 10 **end for**
- 11 Sample batch $\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^b \sim \mathcal{D}$
- 12 $\hat{\mathbf{x}}_i \leftarrow T_\theta(\mathbf{x}_i, \mathcal{T}_c)$
- 13 $\mathcal{L}_{\text{trans}} \leftarrow \frac{1}{b} \sum_{i=1}^b [\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_1 - \text{SSIM}(\mathbf{x}_i, \hat{\mathbf{x}}_i)]$
- 14 $\mathcal{L}_{\text{idt}} \leftarrow \frac{1}{b} \sum_{i=1}^b [\|\mathbf{x}_i - T_\theta(\mathbf{x}_i, \mathcal{T}_h)\|_1 + \|\mathbf{y}_i - T_\theta(\mathbf{y}_i, \mathcal{T}_c)\|_1]$
- 15 $\mathcal{L}_{\text{freq}} \leftarrow 0$
- 16 **for** $i = 1$ to b **do**
- 17 $M_{\hat{\mathbf{x}}} \leftarrow |\mathcal{F}(\hat{\mathbf{x}}_i)|, M_x \leftarrow |\mathcal{F}(\mathbf{x}_i)|, M_y \leftarrow |\mathcal{F}(\mathbf{y}_i)|$
- 18 $\mathcal{L}_{\text{like}} \leftarrow \|M_{\hat{\mathbf{x}}} - M_y\|_F^2, \mathcal{L}_{\text{prior}} \leftarrow -\|M_{\hat{\mathbf{x}}} - M_x\|_F^2$
- 19 $\mathcal{L}_{\text{tv}} \leftarrow TV(\hat{\mathbf{x}}_i)$
- 20 $\mathcal{L}_{\text{freq}} \leftarrow \mathcal{L}_{\text{freq}} + \mathcal{L}_{\text{like}} + \mathcal{L}_{\text{prior}} + \mathcal{L}_{\text{tv}}$
- 21 **end for**
- 22 $\mathcal{L}_{\text{freq}} \leftarrow \frac{\gamma}{b} \cdot \mathcal{L}_{\text{freq}}$
- 23 $\mathcal{L}_{\text{adv}} \leftarrow -\frac{\lambda_{ot}}{b} \sum_{i=1}^b D_\phi(\hat{\mathbf{x}}_i)$
- 24 $\mathcal{L}_G \leftarrow \alpha \mathcal{L}_{\text{trans}} + \beta \mathcal{L}_{\text{idt}} + \mathcal{L}_{\text{freq}} + \mathcal{L}_{\text{adv}}$
- 25 $\theta \leftarrow \theta - \eta_G \cdot \nabla_\theta \mathcal{L}_G$
- 26 **end while**

The complete BFR objective integrates these components as

$$\mathcal{L}_{\text{freq}} = \mathcal{L}_{\text{like}} + \mathcal{L}_{\text{prior}} + \mathcal{L}_{\text{tv}}. \quad (29)$$

Our overall learning objective seeks the OT mapping T^* can be formulated as

$$\begin{aligned} \arg \min_{T_\theta} \sum_{i=1}^m & [\alpha \mathcal{L}_{\text{cost}}(\mathbf{x}_i, T_\theta(\mathbf{x}_i, \mathcal{T}_c)) + \beta \mathcal{L}_{\text{idt}}(\mathbf{y}_i, T_\theta(\mathbf{y}_i, \mathcal{T}_c)) \\ & + \beta \mathcal{L}_{\text{idt}}(\mathbf{x}_i, T_\theta(\mathbf{x}_i, \mathcal{T}_h)) \\ & + \gamma \mathcal{L}_{\text{freq}}(\mathbf{x}_i, \mathbf{y}_i, T_\theta(\mathbf{x}_i, \mathcal{T}_c))] \\ & + \mathcal{W}_1(\mathbb{Q}(\mathcal{Y}), \mathbb{Q}(T_\theta(\mathcal{X}))) \end{aligned} \quad (30)$$

where $\alpha, \beta, \gamma = 1.0$ are balancing weights following [22], [28], [43], $\mathcal{L}_{\text{cost}}(\cdot)$ denotes the domain transport cost, $\mathcal{L}_{\text{idt}}(\cdot)$ is an identity-preserving loss ensuring that inputs from the target domain (specified by text prompt) remain invariant, and $\mathcal{W}_1$ represents the 1-Wasserstein distance between the distribution of clean images $\mathbb{Q}(\mathcal{Y})$ and the distribution of transported hazy images $\mathbb{Q}(T_\theta(\mathcal{X}))$.

E. Overall Pipeline

The overall training procedure of our proposed OBCOT framework is formalized in Algorithm 1. In all experimental configurations, we adopt a one-step stable diffusion model [51] as the backbone network. Our pipeline incorporates three distinct translation mappings parameterized by a shared network $T_\theta, T_\theta(\mathbf{x}, \mathcal{T}_c) : \mathcal{X} \rightarrow \mathcal{Y}, T_\theta(\mathbf{x}, \mathcal{T}_h) : \mathcal{X} \rightarrow \mathcal{X}$, and $T_\theta(\mathbf{y}, \mathcal{T}_c) : \mathcal{Y} \rightarrow \mathcal{Y}$. These mappings utilize identical network architecture but are conditioned on different textual prompts $\mathcal{T}_c$ and $\mathcal{T}_h$, corresponding to distinct reconstruction pathways. As illustrated in Fig. 2, we maintain the majority of pretrained layers frozen, updating only a limited set of zero-initialized convolutional layers and integrated LoRA modules. The learning objective is formulated as a tripartite reconstruction constraint: 1) for a hazy image $\mathbf{x} \in \mathcal{X}$ and the text prompt *clean image*, the model must remove atmospheric degradation while restoring structural and textural details; 2) for a hazy image $\mathbf{x} \in \mathcal{X}$ and the prompt *hazy image*, the model should preserve the input content without introducing artifacts; 3) for a clean image $\mathbf{y} \in \mathcal{Y}$ and the prompt *clean image*, the model must act as an identity mapping, leaving the clean image unaltered. In contrast to conventional diffusion models [67] that initiate generation from random noise, our inference process begins with the latent representation of the input image, obtained via a pretrained variational autoencoder (VAE) encoder. The textual prompts are encoded using a pretrained CLIP text encoder to produce semantic embeddings, which are injected into the intermediate and decoder layers of the pretrained network through cross-attention mechanisms to guide the transformation. The input latent features conditioned on the textual embeddings are processed by the fine-tuned network to yield updated latent representations, which are subsequently reconstructed into the target image via the VAE decoder.

Unlike existing unpaired image dehazing methods [22], [26], [64] that rely only on adversarial training and cycle-consistency losses, our approach formulates the unpaired image dehazing as an OT problem. This formulation provides a more theoretically grounded and stable optimization framework. Meanwhile, our model incorporates a structure-preserving transport cost alongside a Bayesian frequency-domain regularizer, jointly enforcing the structural and spectral consistency between domains. Moreover, the utilization of the pretrained one-step diffusion model, fine-tuned with parameter-efficient LoRA adapters, enables the effective adaptation to the unpaired dehazing task without full model retraining, thereby achieving an advantageous trade-off between the restoration performance and computational efficiency.

IV. EXPERIMENTAL RESULTS**A. Implementation Details**

We conduct experiments using the PyTorch framework on an NVIDIA Tesla A40 GPU. Our proposed model is trained for a total of 200 epochs with a patch size of 256×256 and optimized using the RMSprop optimizer. The initial learning rates are set to 1×10^{-4} for the discriminator and 5×10^{-5}

for the generator, with a decay factor of 10 applied every 100 epochs. During training, the hyperparameters in BFR are set to $\sigma_l = 1.0$, $\sigma_p = 0.5$, and $\lambda_p = 0.1$. These values are derived from a statistical analysis of the frequency-domain characteristics of the training datasets, ensuring they reflect the actual spectral properties of hazy and clean images. In our framework, the distribution-matching term $\mathcal{D}((T_\theta)_\# \mathbb{P}, \mathbb{Q})$ is implemented using an adversarial critic network $D_\phi : \mathcal{Y} \rightarrow \mathbb{R}$, following the WGAN-GP formulation [68]. Specifically, we estimate the 1-Wasserstein distance $\mathcal{D}((T_\theta)_\# \mathbb{P}, \mathbb{Q})$ via

$$\max_{\phi} \left[\mathbb{E}_{\mathbf{y} \sim \mathbb{Q}} [D_\phi(\mathbf{y})] - \mathbb{E}_{\mathbf{x} \sim \mathbb{P}} [D_\phi(T_\theta(\mathbf{x}))] - \lambda_{\text{gp}} \cdot \mathbb{E}_{\hat{\mathbf{y}} \sim \mathbb{P}_{\hat{\mathbf{y}}}} \left[\left(\|\nabla_{\hat{\mathbf{y}}} D_\phi(\hat{\mathbf{y}})\|_2 - 1 \right)^2 \right] \right] \quad (31)$$

where $\mathbb{P}_{\hat{\mathbf{y}}}$ denotes the distribution of linear interpolates between samples from $(T_\theta)_\# \mathbb{P}$ and $\mathbb{Q}$, and λ_{gp} is the gradient penalty coefficient. The overall minimax objective for the transport network T_θ and the critic D_ϕ is

$$\min_{\theta} \max_{\phi} \left\{ \mathbb{E}_{\mathbf{x} \sim \mathbb{P}} [c(\mathbf{x}, T_\theta(\mathbf{x}))] + \lambda_{\text{ot}} \left(\mathbb{E}_{\mathbf{y} \sim \mathbb{Q}} [D_\phi(\mathbf{y})] - \mathbb{E}_{\mathbf{x} \sim \mathbb{P}} [D_\phi(T_\theta(\mathbf{x}))] - \lambda_{\text{gp}} \cdot \mathbb{E}_{\hat{\mathbf{y}} \sim \mathbb{P}_{\hat{\mathbf{y}}}} \left[\left(\|\nabla_{\hat{\mathbf{y}}} D_\phi(\hat{\mathbf{y}})\|_2 - 1 \right)^2 \right] \right) \right\} \quad (32)$$

where $\lambda_{\text{ot}} > 0$ balances the transport cost and distribution alignment. We set $\lambda_{\text{gp}} = 10$, $\lambda_{\text{ot}} = 1.0$, and clip critic gradients to a maximum norm of 0.01 [68]. The critic is updated five times per generator step, and no entropy regularization or Sinkhorn iterations are used, as we directly optimize the 1-Wasserstein distance via the adversarial critic. Furthermore, we randomly shuffle the datasets to ensure that paired images are never presented together, following a commonly used training strategy [22], [27]. The training processes on different datasets are independent. In addition, we define the text prompts in our model as *hazy image* and *clean image* to guide the generation process. During inference, we directly embed the encoded features of both text prompts obtained by the CLIP text encoder into the pretrained one-step diffusion model, reducing the computational overhead.

B. Datasets

We employ Dense-Haze [69], NH-Haze [70], I-HAZE [71], and O-HAZY [71] datasets to evaluate the real-world unpaired image dehazing performance of our proposed model. Specifically, Dense-Haze dataset (45/5/5 pairs for training/validation/testing) is utilized for dense and homogeneous haze removal, NH-Haze dataset (45/5/5 pairs for training/validation/testing) focuses on nonhomogeneous haze, I-HAZE dataset (45/5 pairs for training/testing) is captured under in-door real-world haze condition using a professional haze generation machine, O-HAZY dataset (45/5 pairs for training/testing) is captured under out-door real-world haze condition using a professional haze generation machine.

Meanwhile, to evaluate the effectiveness of our method on real-world unpaired deraining, raindrops removal, and desnowing, we also utilize the RealRain1K [72], LHP-Rain [73], RainDS [74], and WeatherBench [75] datasets to conduct

experiments. Specifically, RealRain1K-H (784/112/224 pairs for training/validation/testing with dense rain streaks) is a sub-dataset of RealRain1K with densely distributed rain streaks. LHP-Rain dataset contains two 100 training pairs, 600 validation pairs, and 300 testing pairs. RainDS dataset [74] in our experiments, the subdataset with degradation of real-world raindrops is employed, which contains 450 image pairs for training and 294 image pairs for evaluation, named RainD-real in our experiments.

C. Evaluation Metrics

We calculate the peak signal-to-noise ratio (PSNR), SSIM, and learned perpetual image patch similarity (LPIPS) [76] between the dehazed images and corresponding ground truth. Meanwhile, we also utilize the language-image quality evaluator (LIQE) [77], natural image assessment (NIMA) [78], multiscale image quality transformer (MUSIQ) [79], and multidimensional attention network image quality assessment (ManIQA) [80] to evaluate the no-reference metrical scores of dehazed images. The lower LPIPS score ($\downarrow$) indicates better image quality, and vice versa for the others ($\uparrow$).

D. Comparisons in Unpaired Dehazing

We compare the performance of our proposed approach with several well-performing existing supervised and unpaired image dehazing methods. The supervised approaches include AODNet [55], GridDehazeNet [56], gUNet-T [57], PhdNet-T [58], and MPMF-Net [59]. The unpaired learning methods contains DCP [10], BCCR [60], CycleGAN [27], DisentGAN [61], ZID [62], YOLY [63], SNSPGAN [64], NLCL [33], UNSB [66], UCL-Dehaze [22], D4 + [21], TAO [65], UME-Net [19], and DehazeSB [54].

As Table I depicted, our proposed method demonstrates substantial enhancements in performance metrics, surpassing existing methods in both reference-based and no-reference evaluations. Specifically, our method achieves performance gains of 0.820, 2.780, 1.948, and 2.226 dB over the second-best unpaired dehazing approaches on Dense-Haze, NH-Haze, I-HAZE, and O-HAZY datasets. Meanwhile, our model also obtains the consistent best performances in terms of photo-realism. Furthermore, even compared to some fully supervised learning methods [55], [58], [59] that employ the paired images for training, our proposed method performs the best in dehazing on all four datasets. To further illustrate the effectiveness of our model, we additionally compute the average quantitative metrics in Table II and offer the intuitive comparisons among various methods across PSNR and four image naturalness assessment metrics in Fig. 1. As illustrated, our model achieves the best performance across the seven metrical indicators, while obtains the best tradeoff between the image fidelity and naturalness. Furthermore, we also provide several visual samples of the comparative methods in Fig. 4. As depicted, the visual results of the other approaches exhibit either notable residual degradation or substantial color distortions. In contrast, our model effectively removes haze degradation while maintaining high-quality content consistency.

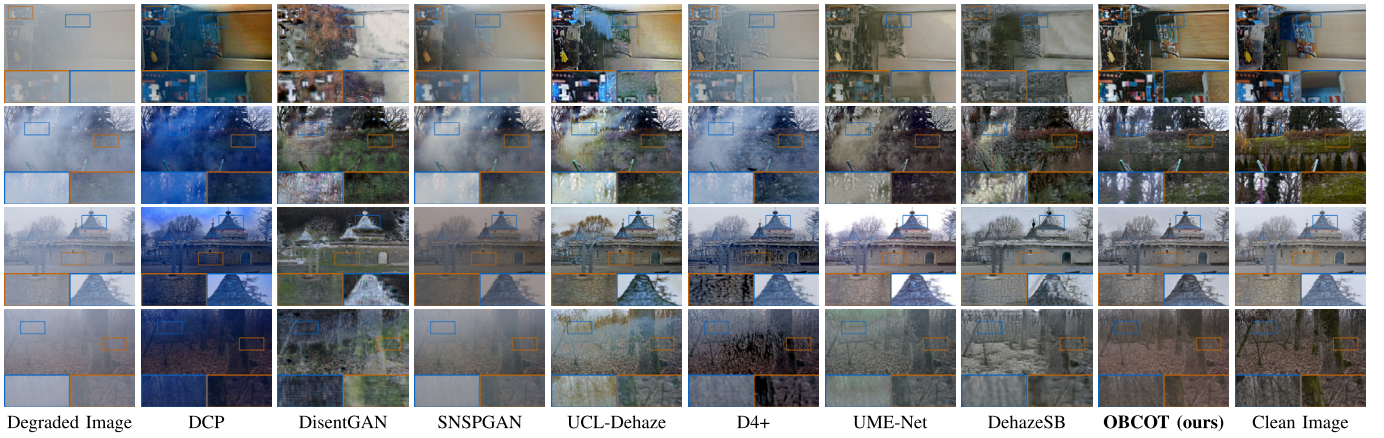


Fig. 4. Visual results of different unpaired learning methods on real-world image dehazing, where the colored boxes denote the zoom-in view patches for better comparisons. Our proposed method achieves the best visual pleasing performance.



Fig. 5. Visual results of different unpaired learning methods on real-world deraining and desnowing, where the colored boxes denote the zoom-in view patches for better comparisons. Our proposed method achieves the best visual pleasing performance.

E. Comparisons in Unpaired Deraining and Desnowing

Furthermore, we evaluate the effectiveness of our proposed method on real-world unpaired deraining, raindrops removal, and desnowing. We set the negative prompt to the corresponding degradation domain description, such as *rainy image*, *snowy image*, and use *clean image* for all positive prompts. We compare the performance of our proposed approach with several well-performing existing supervised and unpaired learning methods. The supervised approaches include JORDER-E [81], PReNet [82], SPANet [83], EfficientDerain [84], and MPMF-Net [59]. The unpaired image deraining methods contains CycleGAN [27], CUT [85], DisentGAN [61], DerainCycleGAN [26], NLCL [33], UNSB [28], CLIP-LIT [86], UCL-Dehaze [22], DestripeCycleGAN [87], NSB-CLIP [23], DehazeSB [54], and RSCP²GAN [88].

As detailed in Table III, fully supervised methods achieve superior performance relative to unpaired deraining approaches in terms of both PSNR and SSIM metrics. However, our proposed method yields substantial performance gains over recent well-performing unpaired deraining approaches across all seven evaluation metrics. Specifically, our model exceeds the second-best method by margins of 0.845 dB on the RealRain1K-H dataset, 0.575 dB on the

LHP-Rain dataset, 1.372 dB on the RainD-real dataset, and 1.234 dB on the WeatherBench-Snow dataset. To further depict the effectiveness of our method for real-world rain and snow removal, visual comparisons of the restored images from several unpaired deraining methods are provided in Fig. 5. The results demonstrate that our method effectively removes the rain degradations and snow particles while preserving the structural and textural details. In contrast, the competing methods exhibit perceptible residual artifacts and pronounced degradation of image textures. Therefore, the proposed approach also demonstrates considerable advantages in both quantitative and qualitative evaluations for real-world deraining and desnowing.

V. ABLATION STUDIES

We conduct several ablation experiments to evaluate the effectiveness of our proposed method. All the metrical scores are calculated and averaged across the four involved testing datasets.

A. Effect of Transport Cost

In Table IV, we evaluate the effectiveness of our structure-preserving transport cost. The results clearly demonstrate that

TABLE I

QUANTITATIVE COMPARISONS OF DIFFERENT UNPAIRED LEARNING METHODS ON REAL-WORLD IMAGE DEHAZING. THE HIGHLIGHTED AND UNDERLINED SCORES ARE THE BEST AND SECOND-BEST, WHILE THE RED AND BLUE FONT DENOTE THE BEST AND SECOND-BEST METRICAL SCORES AMONG THE UNPAIRED LEARNING APPROACHES. OUR PROPOSED METHOD ACHIEVES THE CONSISTENT SUPERIORITY ACROSS THE INVOLVED TESTING DATASETS

Method	Dense-Haze							NH-Haze						
	PSNR (↑)	SSIM (↑)	LPIPS (↓)	LIQE (↑)	NIMA (↑)	MUSIQ (↑)	ManIQA (↑)	PSNR (↑)	SSIM (↑)	LPIPS (↓)	LIQE (↑)	NIMA (↑)	MUSIQ (↑)	ManIQA (↑)
AODNet [55]	13.979	0.4346	0.8289	1.2061	0.3611	35.230	0.2884	14.679	0.4673	0.5049	2.0944	0.5732	59.826	0.3152
GridDehazeNet [56]	15.457	0.4229	0.8799	1.3860	0.2509	22.311	0.2990	16.456	0.5755	0.4669	1.1832	0.5423	52.633	0.2934
gUNet-T [57]	16.846	0.4673	0.7998	1.0837	0.2306	30.357	0.2304	<u>16.885</u>	<u>0.5928</u>	0.4790	2.0540	0.5074	50.255	0.2362
PhDnet-T [58]	16.037	0.4998	0.6720	1.0416	0.2618	29.905	0.2274	15.692	0.5488	0.4370	1.5657	0.5838	51.608	0.2256
MPMF-Net [59]	15.964	0.5084	0.6776	1.2438	0.3471	30.848	0.2662	16.103	0.5827	0.4630	1.8109	0.5492	49.904	0.2569
DCP [10]	10.501	0.4093	0.8122	1.0346	0.3301	35.979	0.2036	11.942	0.4509	0.5775	1.7031	0.5615	60.036	0.3274
BCCR [60]	13.558	0.4562	0.7402	1.0227	0.3761	35.125	0.1871	12.951	0.4841	0.4699	1.9085	0.6136	<u>60.771</u>	0.2995
CycleGAN [27]	13.712	0.3160	0.6267	1.0748	0.5074	48.007	0.2446	15.204	0.3828	0.5362	1.5365	0.6412	51.499	0.3203
DisentGAN [61]	12.270	0.2537	0.7476	1.0100	0.4290	28.138	0.2593	14.287	0.2767	0.6824	1.0147	0.5227	25.635	0.2779
ZID [62]	12.585	0.4424	0.7608	1.0096	0.3412	28.195	0.1458	11.985	0.4047	0.5468	1.5247	0.5851	55.994	0.3037
YOLY [63]	11.059	0.4231	0.7329	1.0922	0.3516	37.720	0.2492	12.638	0.4088	0.5697	1.4848	0.5950	60.518	<u>0.3298</u>
SNSPGAN [64]	12.089	0.4104	0.8361	1.2048	0.3088	25.757	0.2803	12.221	0.4061	0.5961	1.8705	0.5729	60.030	0.3128
NLCL [33]	9.8867	0.2944	0.8517	1.0595	0.4551	42.889	0.2616	12.257	0.4383	0.5147	1.4873	0.6077	52.024	0.2774
UCL-Dehaze [22]	13.343	0.3872	0.6719	1.0021	0.4675	29.166	0.2386	12.767	0.5189	0.5036	1.2695	0.6027	54.890	0.2637
D4+ [21]	10.804	0.3833	0.7094	1.6777	0.3908	36.664	0.2963	12.620	0.4552	0.5528	1.9653	0.5666	59.939	0.3231
TAO [65]	11.350	0.3936	0.8843	1.0130	0.3241	27.836	0.2491	11.303	0.2531	0.8945	1.0140	0.4649	35.839	0.2199
UNSB [66]	14.037	0.3298	0.5720	1.0837	0.4306	50.357	0.2925	14.692	0.4088	0.5370	1.5657	0.5838	41.608	0.2460
NSB-CLIP [23]	14.303	0.3521	0.6004	2.0090	<u>0.6079</u>	<u>60.668</u>	0.3195	14.696	0.4236	0.6245	1.1412	<u>0.6484</u>	47.979	0.2545
UME-Net [19]	<u>15.402</u>	0.4337	<u>0.5308</u>	<u>2.4972</u>	0.6011	59.514	<u>0.3327</u>	14.847	<u>0.5686</u>	<u>0.4172</u>	1.8944	0.6218	54.768	0.3130
DehazeSB [54]	14.146	0.3502	0.8766	1.0113	0.4438	35.454	0.3279	13.941	0.3639	0.6086	1.0500	0.5619	47.110	0.2972
OBCOT (ours)	16.222	0.4639	0.5039	2.8780	0.6237	66.110	0.3420	17.984	0.6372	0.3290	3.0921	0.6783	62.336	0.3608

Method	I-HAZE							O-HAZE						
	PSNR (↑)	SSIM (↑)	LPIPS (↓)	LIQE (↑)	NIMA (↑)	MUSIQ (↑)	ManIQA (↑)	PSNR (↑)	SSIM (↑)	LPIPS (↓)	LIQE (↑)	NIMA (↑)	MUSIQ (↑)	ManIQA (↑)
AODNet [55]	14.852	0.6467	0.4173	1.1192	0.4640	54.509	0.3139	17.765	0.6182	0.3022	1.8621	0.6037	60.269	0.4005
GridDehazeNet [56]	16.598	0.7267	0.3605	1.1798	0.3308	53.196	0.2541	19.377	0.7148	0.2563	2.0539	0.6066	59.850	0.3974
gUNet-T [57]	17.873	0.6947	0.3040	1.0950	0.3092	47.510	0.2099	20.928	0.7311	0.2400	2.0007	0.6279	59.329	0.3612
PhDnet-T [58]	18.948	0.6913	0.2714	1.4245	0.3473	54.375	0.2501	22.491	<u>0.7513</u>	<u>0.1925</u>	2.1828	<u>0.6531</u>	62.011	0.3786
MPMF-Net [59]	17.355	<u>0.7491</u>	0.2732	1.4095	0.3765	55.861	0.2986	<u>21.740</u>	0.7393	0.2105	1.8245	0.6255	60.677	0.3593
DCP [10]	10.829	0.5591	0.3677	1.9545	0.4576	59.700	0.4156	14.867	0.6585	0.3212	1.9201	0.5942	59.601	0.4411
BCCR [60]	14.202	0.7073	0.2817	<u>2.1747</u>	0.4549	61.531	<u>0.4385</u>	15.017	0.6443	0.2778	2.1633	0.6313	59.547	0.4647
CycleGAN [27]	14.615	0.5245	0.4654	1.0138	0.3989	36.456	0.2222	18.776	0.4958	0.3823	1.4246	0.5866	51.757	0.2408
DisentGAN [61]	15.034	0.4115	0.6566	1.0014	0.2261	25.552	0.1403	10.233	0.1003	0.6901	1.0131	0.3761	35.304	0.2496
ZID [62]	13.147	0.6237	0.3635	1.5268	0.4666	56.381	0.381	13.840	0.5159	0.3587	1.5340	0.6359	56.672	0.4070
YOLY [63]	15.291	0.6813	<u>0.2775</u>	1.7013	0.4563	61.462	0.3657	17.865	0.6570	0.2761	1.6563	0.5861	60.620	0.4123
SNSPGAN [64]	15.216	0.6977	0.3145	1.7925	0.4389	59.096	0.3682	17.022	0.6303	0.2988	1.9318	0.5762	58.427	0.3966
NLCL [33]	10.821	0.4854	0.5777	1.2110	0.4344	46.247	0.4008	18.582	0.7350	0.2814	<u>2.2474</u>	0.6455	59.502	<u>0.4667</u>
UCL-Dehaze [22]	15.868	<u>0.7377</u>	0.2914	1.5302	0.3971	56.563	0.2431	17.517	0.7271	0.2828	1.4521	0.6283	58.617	0.3448
D4+ [21]	15.164	0.6167	0.3048	2.0216	0.4232	58.655	0.4076	16.493	0.6404	0.3648	1.3521	0.5215	56.049	0.3929
TAO [65]	15.223	0.6540	0.3944	1.0084	0.3601	46.173	0.2160	16.492	0.4100	0.5983	1.0082	0.4210	44.480	0.2346
UNSB [66]	15.433	0.5482	0.4674	1.0941	0.3987	44.129	0.2025	16.101	0.4311	0.5614	1.0547	0.4471	46.724	0.2345
NSB-CLIP [23]	<u>17.298</u>	0.6823	0.5130	1.0464	0.3144	48.557	0.2132	18.001	0.5591	0.6331	1.1102	0.4706	49.183	0.2468
UME-Net [19]	15.346	0.7274	0.3004	1.5194	<u>0.5170</u>	<u>62.239</u>	0.3934	<u>19.171</u>	<u>0.7371</u>	<u>0.2153</u>	2.0097	<u>0.6499</u>	<u>63.039</u>	0.4529
DehazeSB [54]	15.109	0.5539	0.4729	1.0785	0.5054	54.718	0.1870	16.109	0.4669	0.5836	1.0802	0.4832	55.133	0.2433
OBCOT (ours)	19.246	0.7668	0.1745	3.9760	0.6081	72.459	0.4823	21.397	0.7582	0.1599	2.6689	0.6590	65.103	0.4951

incorporating perceptual loss into the transport cost leads to a moderate performance gain of 0.396 dB in PSNR compared to the baseline without structural constraints. However, the most significant improvement comes from our proposed SSIM-based transport cost, which advances the perceptual loss approach by 1.353 dB in PSNR. This substantial enhancement underscores the superior capability of SSIM in learning the OT mapping between hazy and clean image domains.

B. Effect of Text Guidance

In Table V, we assess the influence of textual guidance by comparing prompts of varying semantic specificity,

fine-grained content text from bootstrapped language-image pretraining (BLIP) model [89] captions, higher level scene prompts, and minimal domain text. As reported, the semantic density of the prompt significantly impacts the stability and performance of the OT mapping. Content prompts, despite being information-rich, lead to the poorest performance in fidelity metrics. We attribute this to training instability and hallucination, where the model is misled by specific content descriptions that may not align perfectly with the input images. Scene prompt demonstrates improved stability and performance over content prompts. However, the domain prompt, with the lowest semantic density, achieves the best results. Their focus purely on the image domain provides a stable

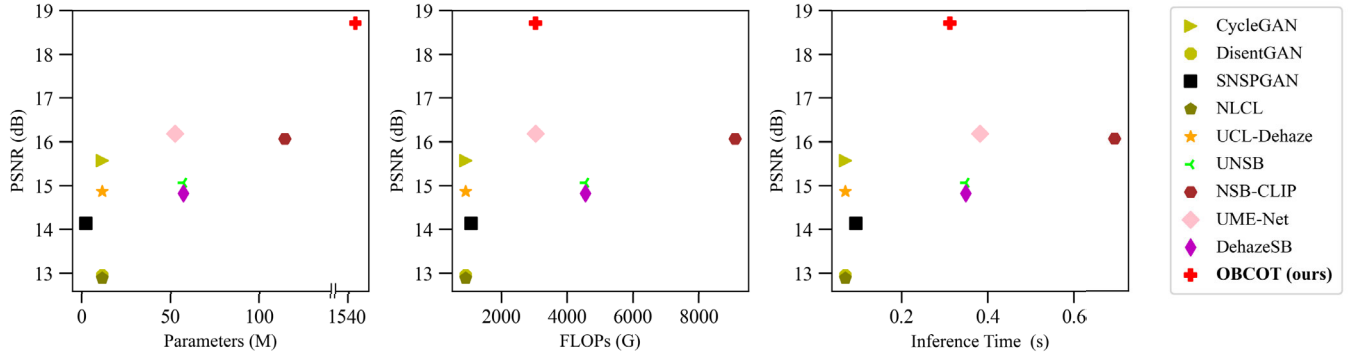


Fig. 6. Efficiency comparisons of our proposed method and the other involved comparative approaches. Our model achieves the best tradeoff between the model efficiency and restoration performance.

TABLE II

AVERAGE QUANTITATIVE COMPARISONS OF DIFFERENT UNPAIRED LEARNING METHODS ON REAL-WORLD IMAGE DEHAZING. THE RED AND BLUE FONT DENOTE THE BEST AND SECOND-BEST METRICAL SCORES. OUR PROPOSED METHOD ACHIEVES THE CONSISTENT SUPERIORITY ACROSS THE INVOLVED TESTING DATASETS

Method	PSNR (↑)	SSIM (↑)	LPIPS (↓)	LIQE (↑)	NIMA (↑)	MUSIQ (↑)	ManIQA (↑)
DCP [10]	12.035	0.5195	0.5197	1.6531	0.4859	53.829	0.3469
BCCR [60]	13.932	0.5730	0.4424	1.8173	0.5190	54.244	0.3475
CycleGAN [27]	15.577	0.4298	0.5026	1.2624	0.5335	46.930	0.2570
DisentGAN [61]	12.956	0.2606	0.6942	1.0098	0.3885	28.657	0.2318
ZID [62]	12.889	0.4967	0.5074	1.3988	0.5072	49.310	0.3094
YOLY [63]	14.213	0.5426	0.4640	1.4836	0.4973	55.080	0.3393
SNSPGAN [64]	14.137	0.5361	0.5114	1.6999	0.4742	50.828	0.3395
NLCL [33]	12.887	0.4883	0.5564	1.5013	0.5357	50.166	0.3516
UCL-Dehaze [22]	14.874	0.5927	0.4374	1.3135	0.5239	49.809	0.2726
D4+ [21]	13.770	0.5239	0.4829	1.7542	0.4755	52.827	0.3550
TAO [65]	13.592	0.4277	0.6929	1.0109	0.3925	38.582	0.2299
UNSB [66]	15.066	0.4295	0.5345	1.1996	0.4651	45.705	0.2439
NSB-CLIP [23]	16.074	0.5043	0.5928	1.3267	0.5103	51.597	0.2585
UME-Net [19]	16.192	0.6167	0.3659	1.9802	0.5975	59.890	0.3730
DehazeSB [54]	14.826	0.4337	0.6354	1.0550	0.4986	48.104	0.2639
OBCOT (ours)	18.712	0.6565	0.2918	3.1538	0.6423	66.502	0.4201

and consistent learning signal that perfectly complements the distribution-alignment goal of OT, preventing distraction by extraneous semantic information and facilitating a more robust domain transfer. The baseline w/o Text performs reasonably but is outperformed by both domain and scene-attribute prompts. This experiment highlights the critical importance of domain-aligned textual guidance in unpaired image dehazing.

1) *Impact of BFR*: Table VI presents the analysis between our proposed BFR and adversarial loss $\mathcal{L}_{adv}$. As depicted, BFR achieves a performance improvement of 1.204 dB in PSNR and 22.15% in NIMA, surpassing adversarial training in unpaired deraining. Therefore, this result validates the effectiveness of our BFR in unpaired image dehazing.

2) *Effect of Individual Components*: Table VII evaluates the contributions of three core key components in our model, the pretrained network, the LoRA adapters, and the zero-convolution layers. The results demonstrate that the pre-trained one-step diffusion model serves as the foundational component. Its removal leads to the most severe decreases across all metrics, confirming the powerful generative priors it provides are indispensable for unpaired image dehazing.

The importance of efficient fine-tuning is highlighted by the results for the LoRA adapters. Discarding the LoRA adapters causes a significant performance decline, notably a 1.067 dB reduction in PSNR. This indicates the critical role of LoRA in effectively adapting the general pretrained model to the specific unpaired dehazing without compromising its inherent capabilities. The zero convolution layers act as a structural anchor, providing a clear benefit to pixel-level fidelity (PSNR and SSIM). However, this strong structural constraint appears to slightly suppress the model flexibility in generating textures with the highest perceived naturalness (LIQE and NIMA). Incorporating the zero convolution in our final configuration results in the most balanced and effective overall performance between the pixel-level fidelity and perceptual realism.

VI. EFFICIENCY ANALYSIS

We present the efficiency comparisons of various methods in Table VIII. Notably, CycleGAN [27], DisentGAN [61], NLCL [90], and UCL-Dehaze [22] share the same generator architecture, resulting in identical parameter counts (#Param), floating operations (FLOPs), and inference times. To evaluate the inference speed, we process 100 degraded images at a resolution of 1024×1024 and report the average inference time. While our proposed method demands more parameters than existing approaches, it reduces the computational burden (see the subbox) by 33.238% compared to recent well-performing UNSB [28] and DehazeSB [54], respectively. Regarding the inference time, existing methods with lightweight architectures achieve the fastest performance. However, by leveraging a pretrained one-step stable diffusion model, our model is faster than several modern methods. Specifically, it reduces the inference time by 55.024% compared to NSB-CLIP [23] and by 10.384% compared to DehazeSB [54]. In summary, although our proposed method requires significantly more parameters than existing approaches, it achieves a competitive inference speed and lower computational cost than several recent models. Furthermore, we visualize the results of Table VIII in Fig. 6, demonstrating that our model achieves a balanced efficiency among existing methods while maintaining high-quality unpaired image restoration.

VII. LIMITATIONS

Although our model exhibits substantial advancements in fidelity and photo-realism compared to existing unpaired

TABLE III

QUANTITATIVE COMPARISONS OF DIFFERENT UNPAIRED LEARNING METHODS ON REAL-WORLD IMAGE DERAISING AND DESNOWING. THE HIGHLIGHTED AND UNDERLINED SCORES ARE THE BEST AND SECOND-BEST, WHILE THE RED AND BLUE FONT DENOTE THE BEST AND SECOND-BEST METRICAL SCORES AMONG THE UNPAIRED LEARNING APPROACHES. OUR PROPOSED METHOD ACHIEVES THE CONSISTENT SUPERIORITY ACROSS THE INVOLVED TESTING DATASETS

Method	RealRain1K-H							LHP-Rain						
	PSNR (↑)	SSIM (↑)	LPIPS (↓)	LIQE (↑)	NIMA (↑)	MUSIQ (↑)	ManIQA (↑)	PSNR (↑)	SSIM (↑)	LPIPS (↓)	LIQE (↑)	NIMA (↑)	MUSIQ (↑)	ManIQA (↑)
JORDER-E [81]	29.843	0.9064	0.2569	1.0403	0.2112	21.359	0.2047	31.240	0.9157	0.1282	3.2819	0.6236	63.489	0.3609
PRerNet [82]	<u>30.943</u>	<u>0.9364</u>	<u>0.1869</u>	1.1638	0.2740	25.500	0.2311	<u>33.340</u>	0.9364	0.0982	3.4818	0.6436	67.489	0.4043
SPANet [83]	30.754	0.9382	0.2025	1.1403	0.2312	22.359	0.2147	<u>33.205</u>	<u>0.9350</u>	0.1243	3.4920	0.6642	65.831	0.4200
EfficientDerain [84]	29.027	0.9116	0.2979	1.0615	0.2257	20.775	0.1987	32.363	0.9253	0.0844	3.4529	0.6567	66.827	0.3887
MPMF-Net [59]	31.832	0.9286	0.1215	1.2322	0.2769	26.421	0.2929	34.272	0.9196	0.0614	3.6804	0.5975	62.190	0.4155
CyCleGAN [27]	20.907	0.7906	<u>0.3858</u>	1.1894	0.3264	30.008	0.2064	28.946	0.9043	0.1117	3.6295	0.6618	67.669	0.4210
CUT [85]	19.663	0.7985	0.4590	1.2779	0.3163	30.611	0.1967	28.398	0.9011	0.1205	3.7039	0.6684	<u>69.018</u>	0.4474
DisentGAN [61]	20.451	0.7725	0.3978	1.1656	0.3195	31.358	0.2019	28.326	0.8846	0.1294	3.5422	0.6472	68.174	0.4215
DerainCycleGAN [26]	23.212	0.7983	0.5053	1.0772	0.3199	31.028	0.2097	31.321	0.9116	0.0990	3.6240	0.6646	68.276	0.4468
DCD-GAN [32]	10.847	0.5949	0.6320	1.0602	0.2856	26.773	<u>0.2418</u>	26.371	0.8426	0.1852	2.8810	0.6234	65.552	0.2947
NLCL [33]	19.537	0.7844	0.4892	1.2577	0.3267	32.616	0.1796	20.807	0.8258	0.1984	3.1432	0.6401	62.171	0.3629
UNSB [28]	17.613	0.7517	0.5021	1.0936	0.3376	32.835	0.2162	23.437	0.7135	0.2830	2.2354	0.6442	66.068	0.2900
CLIP-LIT [86]	22.343	0.7137	0.5285	1.0743	0.3233	30.352	0.2063	28.107	0.9122	0.0999	3.2450	0.6305	64.780	0.4168
UCL-Dehaze [22]	21.343	<u>0.8087</u>	0.4516	<u>1.3245</u>	0.3429	31.548	0.1963	30.499	<u>0.9161</u>	0.0967	<u>3.8198</u>	0.6684	68.810	<u>0.4590</u>
DestripeCycleGAN [87]	23.415	0.7124	0.5304	1.0655	0.3935	<u>34.745</u>	0.2395	29.810	0.8993	0.1492	3.2482	0.6520	67.761	0.4497
NSB-CLIP [23]	<u>24.337</u>	0.7988	0.4604	1.1861	0.3529	33.463	0.2124	<u>32.235</u>	0.9072	<u>0.0936</u>	3.2396	0.6573	67.781	0.4504
DehazeSB [54]	20.253	0.7561	0.4920	1.1727	0.3308	32.163	0.2118	27.956	0.8992	0.1373	3.0907	0.6513	66.746	0.4242
RSCP ² GAN [88]	23.763	0.8023	0.5152	1.1556	0.3235	32.207	0.2055	30.694	0.8934	0.0970	3.5515	0.6513	66.910	0.4379
OBCOT (ours)	25.182	0.8844	0.3004	1.3345	<u>0.3847</u>	36.358	<u>0.2672</u>	32.810	0.9306	<u>0.0794</u>	3.8326	<u>0.6674</u>	69.880	0.4797

Method	RainD-real							WeatherBench-Snow						
	PSNR (↑)	SSIM (↑)	LPIPS (↓)	LIQE (↑)	NIMA (↑)	MUSIQ (↑)	ManIQA (↑)	PSNR (↑)	SSIM (↑)	LPIPS (↓)	LIQE (↑)	NIMA (↑)	MUSIQ (↑)	ManIQA (↑)
JORDER-E [81]	18.343	0.5977	0.3873	1.3156	0.4783	40.299	0.1686	24.626	0.8052	0.2293	1.2364	0.4215	42.392	0.2427
PRerNet [82]	19.308	<u>0.6292</u>	0.4077	1.5465	0.5901	46.223	0.2231	25.712	0.8330	0.2011	1.2491	0.4709	46.040	0.2750
SPANet [83]	19.415	<u>0.6083</u>	0.3821	1.2156	0.6083	47.234	0.2386	<u>25.841</u>	<u>0.8357</u>	0.2097	1.3061	0.5018	44.815	0.2522
EfficientDerain [84]	18.681	0.5365	0.6210	1.0140	0.3639	32.331	0.1418	24.503	0.7813	0.2537	1.2053	0.4016	41.632	0.2258
MPMF-Net [59]	20.827	0.6545	<u>0.2756</u>	1.6218	0.6342	47.492	0.2406	26.123	0.8404	0.1563	1.3519	0.5317	48.912	<u>0.2785</u>
CyCleGAN [27]	19.024	0.5836	0.3036	1.7524	0.5971	46.930	0.2346	22.728	0.7636	0.2308	1.3799	0.5073	47.483	0.2241
CUT [85]	18.525	0.5698	0.3245	1.4999	0.6061	44.752	0.2242	22.109	0.7424	0.2443	1.3386	0.4825	45.054	0.2202
DisentGAN [61]	17.832	0.5516	0.3388	1.5257	0.5708	44.887	0.2281	21.819	0.7494	0.2822	1.4119	0.5307	48.862	0.2135
DerainCycleGAN [26]	19.033	<u>0.5926</u>	0.4357	1.5347	0.6011	45.202	0.2392	22.649	0.7644	<u>0.2261</u>	1.3737	0.4979	46.409	0.2198
DCD-GAN [32]	18.636	0.5866	0.3835	1.6479	0.6096	47.231	0.2341	19.750	0.7686	0.3518	1.8985	<u>0.5517</u>	<u>51.762</u>	0.2524
NLCL [33]	17.308	0.4814	0.4135	1.6416	0.6115	41.933	0.2186	18.930	0.6713	0.3011	1.3020	0.4771	46.121	0.2110
UNSB [28]	17.242	0.5358	0.5384	1.4830	<u>0.6475</u>	47.500	0.2413	17.687	0.5805	0.3881	1.8582	0.5304	49.650	0.2549
CLIP-LIT [86]	18.834	0.5845	<u>0.2902</u>	1.7701	0.5671	43.256	0.2215	21.550	<u>0.7767</u>	0.2412	1.2532	0.4639	45.335	0.2184
UCL-Dehaze [22]	18.532	0.5646	0.3108	1.6057	0.5838	42.587	0.2061	21.719	0.7494	0.2522	1.4119	0.5417	48.762	0.2220
DestripeCycleGAN [87]	18.951	0.5842	0.2920	<u>1.7760</u>	0.5684	43.381	0.2201	21.998	0.7407	0.2495	1.3313	0.4737	44.018	0.2131
NSB-CLIP [23]	18.765	0.5804	0.6784	1.4432	0.6015	42.905	<u>0.2585</u>	22.458	0.7643	0.2286	1.3678	0.5069	47.366	0.2208
DehazeSB [54]	17.180	0.5407	0.4854	1.4913	0.6218	49.896	0.2264	19.858	0.6992	0.2815	1.8042	0.5254	50.024	<u>0.2624</u>
RSCP ² GAN [88]	<u>19.162</u>	0.5853	0.3339	1.5910	0.5847	43.876	0.2322	<u>22.848</u>	0.7640	0.2289	1.4265	0.5152	48.669	0.2286
OBCOT (ours)	<u>20.534</u>	<u>0.6066</u>	0.2581	2.0042	0.6477	<u>49.276</u>	0.2607	<u>24.082</u>	<u>0.8187</u>	<u>0.1905</u>	<u>1.8724</u>	0.5828	58.452	0.2914

TABLE IV

ABLATION EXPERIMENTS ON THE TRANSPORT COST. OUR PROPOSED STRUCTURE-PRESERVING TRANSPORT COST PRESERVES THE DETAILS OF GENERATED IMAGES

Component	PSNR (↑)	SSIM (↑)	LIQE (↑)	NIMA (↑)
w/o SSIM	16.963	0.6012	2.8955	0.6281
w/ Perpetual Loss	<u>17.359</u>	<u>0.6240</u>	<u>3.0604</u>	<u>0.6475</u>
w/ SSIM	18.712	0.6565	3.1538	0.6423

TABLE V

ABLATION EXPERIMENTS ON THE TEXT GUIDANCE. OUR DOMAIN-REPRESENTATION PROMPTS FACILITATE THE PROPOSED MODEL TO ACHIEVE A SIGNIFICANT PERFORMANCE IMPROVEMENT

Text	PSNR (↑)	SSIM (↑)	LIQE (↑)	NIMA (↑)
w/o Text	<u>17.485</u>	<u>0.6280</u>	2.7444	0.6025
w/ Content Prompt	13.205	0.4345	2.9082	<u>0.6372</u>
w/ Scene Prompt	16.983	0.6223	<u>2.9341</u>	0.6184
w/ Domain Prompt	18.712	0.6565	3.1538	0.6423

dehazing methods, it is subject to several limitations. First, despite algorithmic optimization emphasizing structural preservation and frequency regularization, the generalizability of our model across different weather conditions remains limited. Second, when evaluated on real-world hazy images

under extreme conditions, we observe three types of failures. As Fig. 7 illustrated, in scenes with dense atmospheric interference or challenging lighting conditions, the model occasionally introduces color shifts or residual degradation.

TABLE VI
ABLATION EXPERIMENTS ON THE BFR. OUR PROPOSED
APPROACH OBTAINS THE BEST PERFORMANCE

Component	PSNR ($\uparrow$)	SSIM ($\uparrow$)	LIQE ($\uparrow$)	NIMA ($\uparrow$)
$\mathcal{L}_{adv}$	17.508	0.6344	2.5820	0.6005
BFR	18.712	0.6565	3.1538	0.6423

TABLE VII
ABLATION EXPERIMENTS ON THE INDIVIDUAL MODULES. THE LORA
ADAPTERS AND ZERO-CONVOLUTION LAYERS FACILITATE
THE PROPOSED MODEL TO ACHIEVE A SIGNIFICANT
PERFORMANCE IMPROVEMENT

Module	PSNR ($\uparrow$)	SSIM ($\uparrow$)	LIQE ($\uparrow$)	NIMA ($\uparrow$)
w/o Pre-trained	15.232	0.5576	2.3011	0.5512
w/o LoRA Adapters	17.645	0.6248	2.8944	0.6088
w/o Zero Convolution	16.602	0.5881	3.1740	0.6405
OBCOT (ours)	18.712	0.6565	3.1538	0.6423

TABLE VIII
EFFICIENCY COMPARISONS OF DIFFERENT UNPAIRED DEHAZING
METHODS. THE HIGHLIGHTED AND UNDERLINED FONT
DENOTE THE BEST AND SECOND-BEST METRICAL
SCORES AMONG THE UNPAIRED LEARNING APPROACHES
IN THE SUBBOX. OUR PROPOSED METHOD ACHIEVES
A COMPETITIVE BALANCE BETWEEN COMPUTATIONAL
COMPLEXITY AND INFERENCE SPEED

Method	#Param ($\downarrow$)	FLOPs ($\downarrow$)	Inference Time ($\downarrow$)
CycleGAN [27]	11.378 M	909.83 G	0.0695 s
DisentGAN [61]	11.378 M	909.83 G	0.0695 s
SNSPGAN [64]	2.3580 M	1074.3 G	0.0941 s
NLCL [90]	11.378 M	909.83 G	0.0695 s
UCL-Dehaze [22]	11.378 M	909.83 G	0.0695 s
UNSB [28]	56.891 M	4549.1 G	0.3486 s
NSB-CLIP [23]	113.84 M	9098.3 G	0.6946 s
UME-Net [19]	52.228 M	3033.9 G	0.3816 s
DehazeSB [54]	56.891 M	4549.1 G	0.3486 s
OBCOT (ours)	1540.3 M	3037.2 G	0.3124 s



Fig. 7. Failure instances of our proposed method for the haze removal under extreme conditions. The top row displays the degraded input images, while the bottom row presents the reconstructed outputs generated by our model.

VIII. CONCLUSION

In this work, we investigate the effectiveness of treating haze removal as an OT problem. We propose a novel OBCOT to achieve photo-realistic haze removal. Extensive experiments on benchmark datasets demonstrate that our approach significantly enhances both image fidelity and naturalness compared to existing state-of-the-art unpaired dehazing methods. In the future, we will investigate strategies for decoupling degradations from image content and explore prompt-invariant transport mechanisms to enhance cross-domain robustness.

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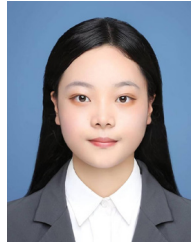
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